



Mentor: Prof. Sanket

Team Members: Manish Mallapur, Ronit Raj, Vaishnavi Peddabudhi, Victor Mukunga,
Leah Daisy, Sai Satyam Mohapatra, Swapnendu Narayan, Allan Chang
Ishanya (Team Lead)

Details of the team

- Prof. Sanket, BITS, India

Team members

- Manish Mallapur, IISER Bhopal, India
- Ronit Raj, IISER Bhopal, India
- Vaishnavi Peddabudhi, India
- Victor Mukunga, Kenyatta University, Kenya
- Leah Daisy, Co-operative University,
- Sai Satyam Mohapatra, IISER Bhopal, India
- Swapnendu Narayan Ghosh, IIT Delhi, India
- Allan Chang, Simon Fraser University, Canada

Team Lead: Ishanya, IISER Bhopal

Device details

Working Principle of Tip-Tilt MEMS Mirrors

Tip-tilt MEMS mirrors control light beams by tilting their reflective surface in two orthogonal directions. This motion is achieved using:

- **Electrostatic Actuation:** Low-power, scalable motion via forces generated between the mirror and electrodes.
- **Piezoelectric Actuation:** High precision and rapid response through electric-field-induced material deformation.
- **Electromagnetic Actuation:** High-force motion using magnetic fields.

Project Focus

This project highlights **electrostatic actuation**, where applied voltages create forces to tilt the mirror. The tilt angle, determined by voltage magnitude and configuration, enables precise, dynamic light redirection.

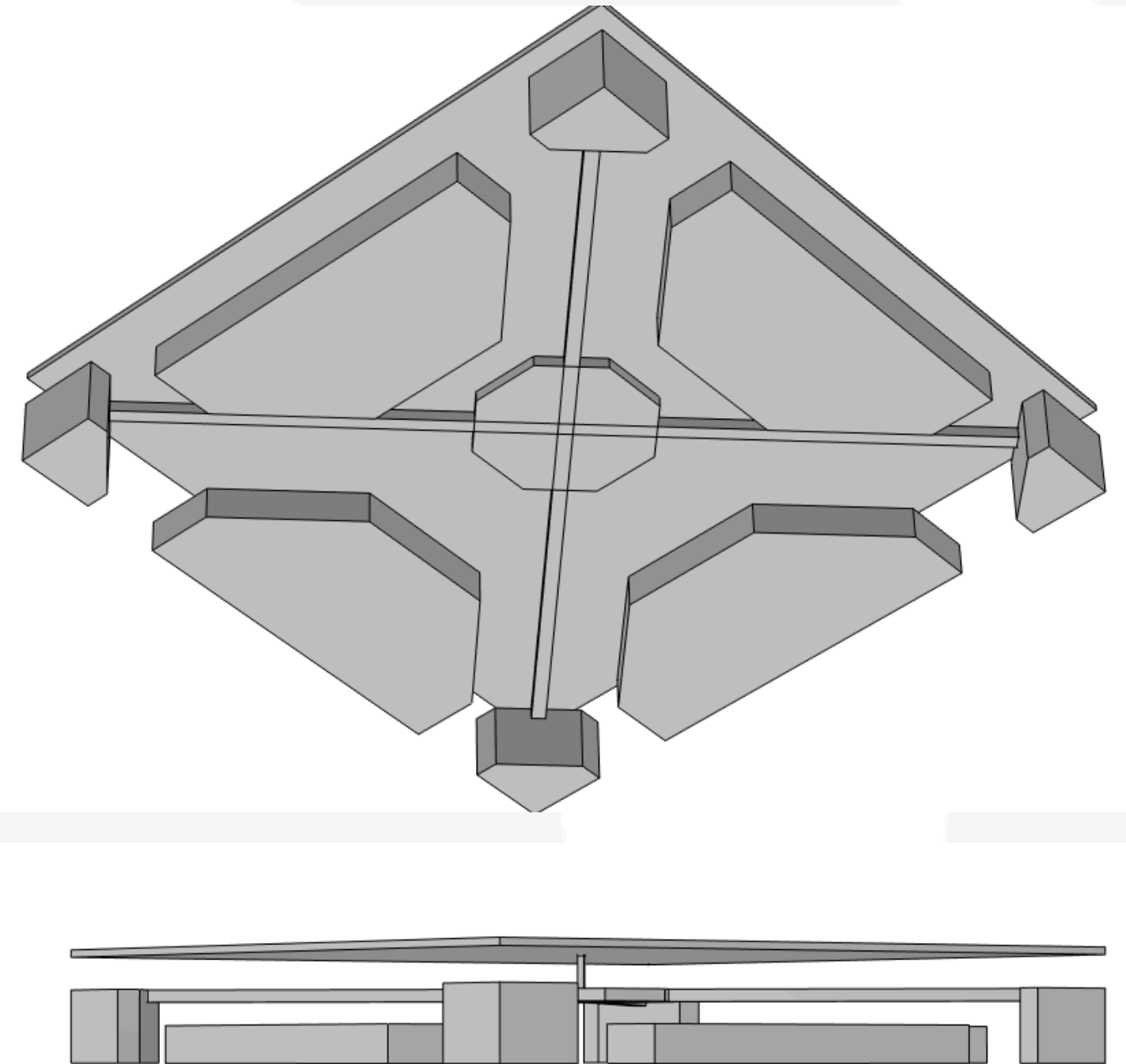
Micromirror (Electrostatic Actuator)

Device dimensions:

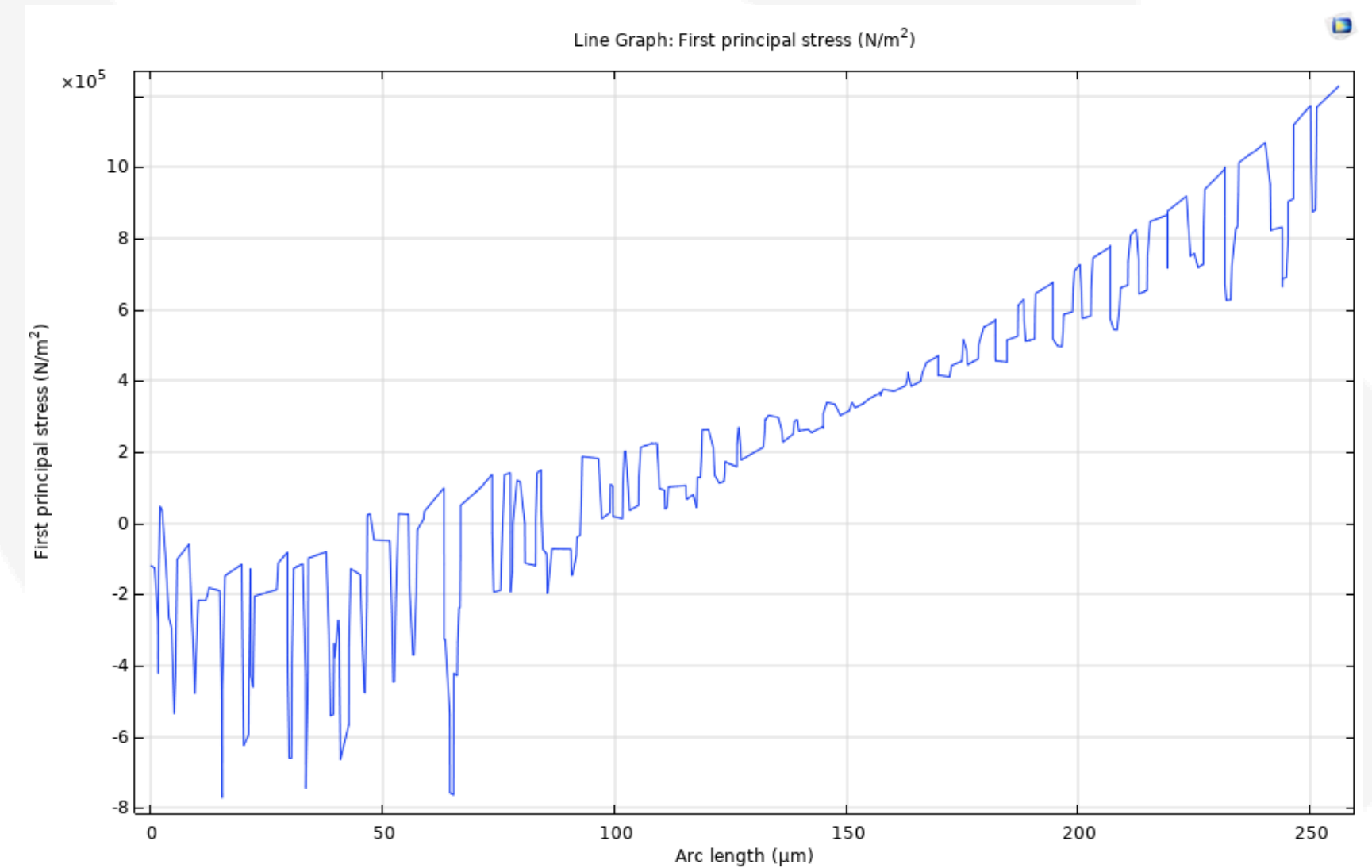
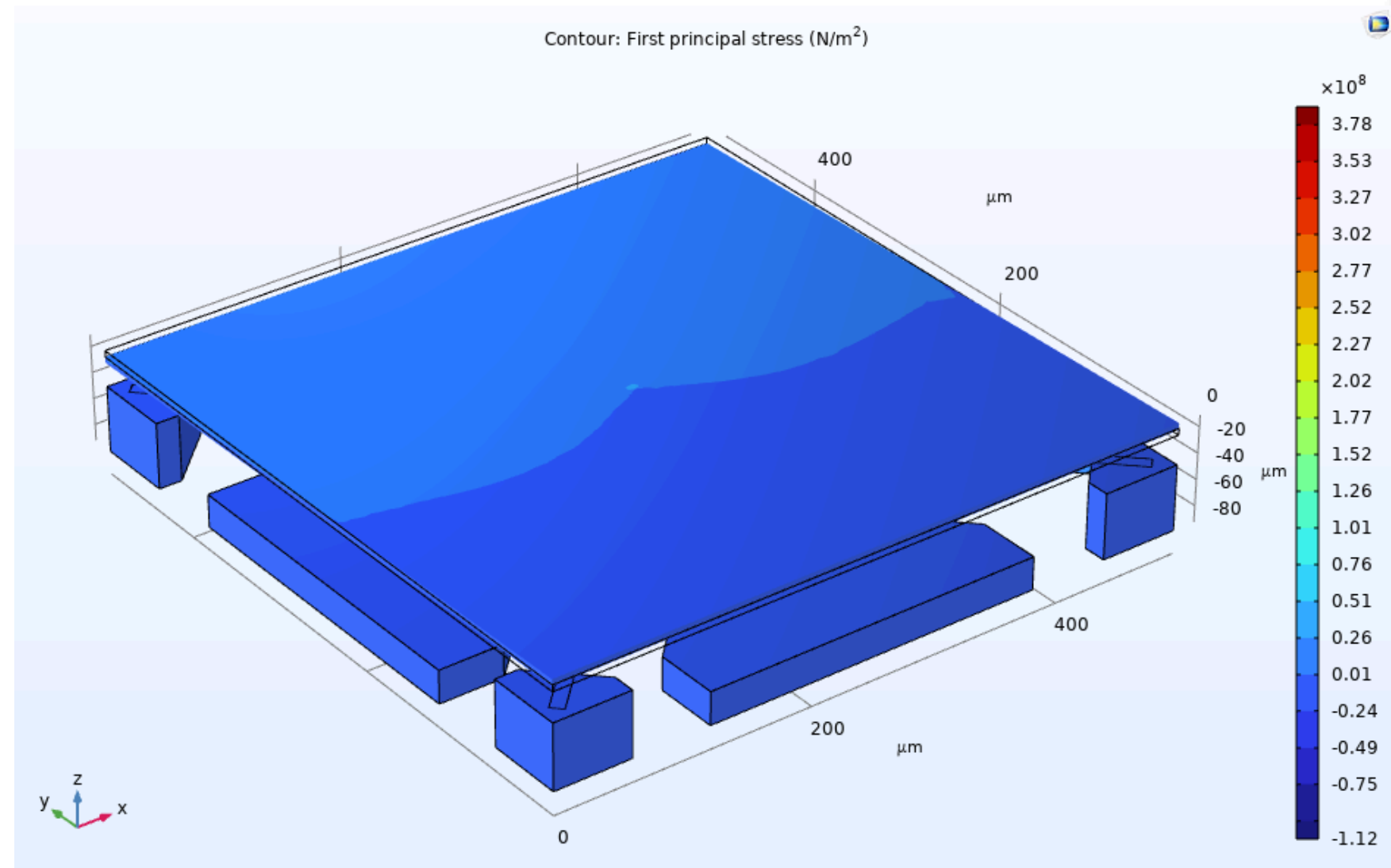
- Length: 500 μm
- Breadth: 500 μm
- Height: 100 μm

Design Approach:

The tip-tilt MEMS sensor features a square mirror base supported by a central cylindrical structure and anchored by an octagonal base with cantilever beams for stability. Four gold electrodes around the square base enable electrostatic actuation, tilting the mirror using silicon for structure and gold for conductivity.

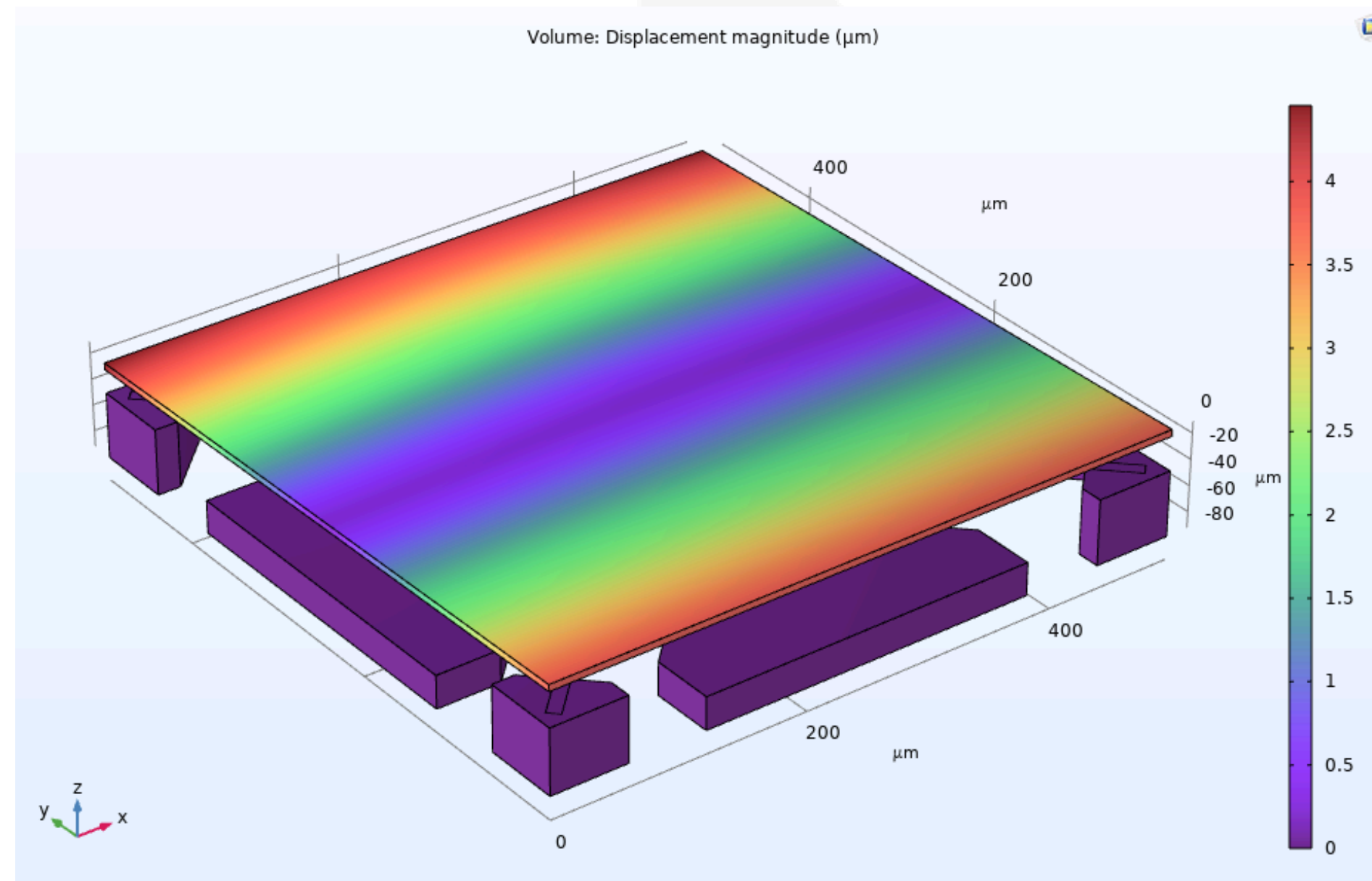


Stress plot



Displacement Plot

On applying a potential on one of the electrodes



Thank you!